magex

Magnetic exchange, anisotropy and magnons from Wannier Hamiltonians

Green's-function theory · full derivations · code walkthrough · validation

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November 10, 2026



Roadmap

1. Goal: the extended Heisenberg model
2. Why a new code: limits of LKAG/TB2J
3. The tight-binding model and its spin decomposition
4. Magnetic force theorem & Lloyd formula
5. Rigid rotations, Hessians, Green's functions
6. Energy integration with Ozaki poles
7. Rotational invariance: partitioning the exchange field
8. Mapping to J, D, J^{ani}, A from one DFT run
9. Symmetrisation and supercell aliasing
10. Beyond MFT: Bruno renormalisation, ligand downfolding
11. Finite angles: Lloyd pair energies, biquadratic terms
12. Linear spin-wave theory
13. MPI implementation
14. Validation and physics results ($SrMnO_3$, bcc Fe, CrI_3)

Colour code used throughout

colour	quantity	symbols
green	Hamiltonian, exchange field	H, H_0, X, p_i, Y
orange	spin–orbit coupling	S, $\xi\,\mathrm{L}\cdot\mathrm{S}$
blue	Green's function, eigenstates	G, Ψ, $\varepsilon_{n\mathrm{k}}$, U_a
red	spin-model parameters	J, D, J^{ani}, A
purple	rotations and directions	e_i, n_i, ϕ, $\mathcal{R}$, θ
teal	Hessians, perturbations	K_{ab}, T_i, W_{ia}, D_i
brown	energies, occupation, contour	Ω, f, μ, z_p, w_p
gold	magnons	$\omega(\mathrm{q})$, b_i, $h(\mathrm{k})$
`slate`	code / MPI	`engine.py`, `Allreduce`

Goal and motivation

The target: an extended Heisenberg model

Spin Hamiltonian (TB2J convention)

$$E[\{\mathbf{e}\}] = \sum_i \mathbf{e}_i \cdot A_i \cdot \mathbf{e}_i \; - \sum_{i \neq j} \mathbf{e}_i \cdot J_{ij} \cdot \mathbf{e}_j$$

unit spins $|\mathbf{e}_i| = 1$; every ordered pair counted, $J_{ji} = J_{ij}^T$; $J > 0$ ferromagnetic.

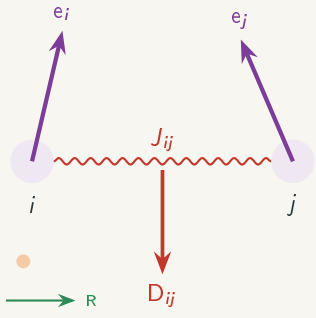
Decomposition of the 3×3 tensor

$$J_{ij} = \underbrace{J_{ij}^{\mathrm{iso}} \mathbb{1}}_{\text{Heisenberg}} + \underbrace{J_{ij}^{\mathrm{ani}}}_{\text{sym., traceless}} + \underbrace{J_{ij}^{A}}_{\text{antisym.}}$$

$$-\mathbf{e}_i \cdot J_{ij}^{A} \cdot \mathbf{e}_j = -\mathbf{D}_{ij} \cdot (\mathbf{e}_i \times \mathbf{e}_j), \quad \mathbf{D} = (J_{yz}^{A}, J_{zx}^{A}, J_{xy}^{A})$$

isotropic ⇒ collinear

DMI ⇒ canting, skyrmions



Take-away: every quantity the code prints is a coefficient of this energy function; the job is to extract all of them from one DFT Hamiltonian.

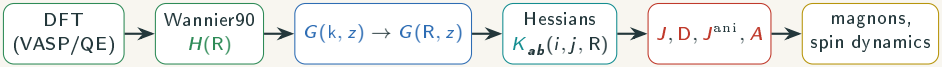
Two ways to get the parameters

Energy mapping

- DFT for many spin configurations, large supercells
- fit J to total energies
- metallisation of high-energy states, supercell-dependent fits, many non-collinear SOC runs

Green's function / force theorem (LKAG)

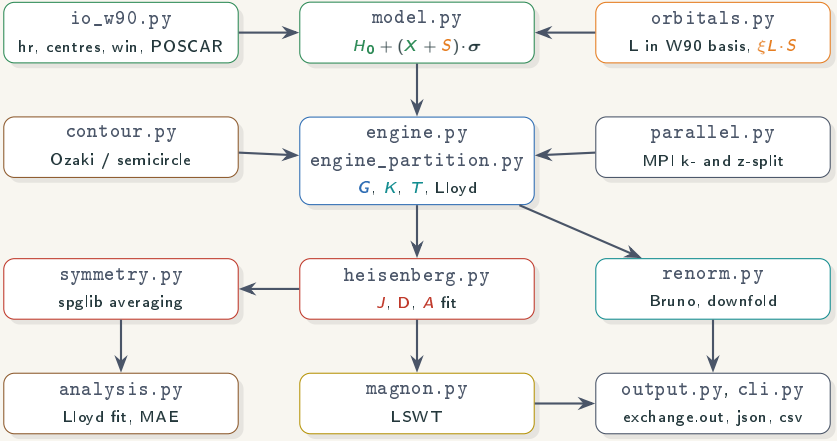
- one ground-state calculation
- J_{ij} at all distances from $G_{ij}(\mathrm{R})$
- local linear response $\Rightarrow$ exact small-angle curvature of the band energy



What limits the standard LKAG/TB2J implementation

limitation	magex
only the *on-site* exchange block is rotated; inter-atomic spin-dependent Wannier elements stay fixed $\Rightarrow$ no global rotation, Goldstone sum rule broken	weighted partition of the full $X(\mathrm{R})$; sum rule exact; bcc Fe J_1: 9.5 $\rightarrow$ 21.2 meV
full tensor needs 3 DFT runs (x, y, z) and a merge	rotate X relative to the lattice in one Hamiltonian
no single-ion anisotropy from Wannier input	A_i from on-site Hessians + band MAE cross-check
$T = 0$ contour with user E_{min}	Ozaki poles at finite T, no E_{min}
plain MFT	Bruno renormalisation
ligand moments rigid/ignored	Schur-complement downfolding
biquadratic from small angles "often unphysical"	exact Lloyd finite-angle pair energies
Wannier noise unquantified	spglib symmetrisation; change reported
magnons isotropic only	LSWT with full J, D, A

Code architecture



driver.py orchestrates; ∼2300 lines Python, numpy/scipy, mpi4py, spglib

The tight-binding model

Real space → Bloch

$$H_{mn}(\mathrm{R}) = \langle w_{m\mathbf{0}} | \hat{H} | w_{n\mathrm{R}} \rangle \quad \Longrightarrow \quad H(\mathrm{k}) = \sum_{\mathrm{R}} \frac{H(\mathrm{R})}{N_{\mathrm{deg}}(\mathrm{R})}\, e^{2\pi i \mathrm{k}\cdot\mathrm{R}}$$

- `read_hr`: header, Wigner–Seitz degeneracies N_{deg} divided out once.
- `read_win`: `unit_cell_cart`, `atoms_frac/cart`, `projections` (s, p, d, f or $l=$), Fortran `1.0d0` floats; `read_poscar` if the .win has no structure.
- Projections are expanded in Wannier90 order: line → site → m; each WF gets (atom, l, m).
- Spinors: `interleaved` (QE, VASP+W90: orbital-major) or `blocked`; internally always spin-block order [↑ all orbitals | ↓ all orbitals].

Collinear input (ISPIN=2)

$$H_0 = \tfrac{1}{2}(H^{\uparrow} + H^{\downarrow}), \qquad X_z = \tfrac{1}{2}(H^{\uparrow} - H^{\downarrow}), \qquad X_{x,y} = S = 0$$

Spin decomposition of a spinor Hamiltonian

Every model is stored as

$$H(\mathrm{R}) = H_0(\mathrm{R}) \otimes \mathbb{1} + \sum_{c=x,y,z} \left[X_c(\mathrm{R}) + S_c(\mathrm{R})\right] \otimes \sigma_c$$

Inverting the Pauli algebra on the 2×2 spin blocks:

$$H_0 = \tfrac{1}{2}(H_{\uparrow\uparrow} + H_{\downarrow\downarrow}), \quad h_z = \tfrac{1}{2}(H_{\uparrow\uparrow} - H_{\downarrow\downarrow}),$$
$$h_x = \tfrac{1}{2}(H_{\uparrow\downarrow} + H_{\downarrow\uparrow}), \quad h_y = \tfrac{i}{2}(H_{\uparrow\downarrow} - H_{\downarrow\uparrow}).$$

since $\sigma_x \to (\uparrow\downarrow, \downarrow\uparrow) = (1, 1)$, $\sigma_y \to (-i, i)$.

	$\uparrow$	$\downarrow$
$\uparrow$	$H_0 + h_z$	$h_x - ih_y$
$\downarrow$	$h_x + ih_y$	$H_0 - h_z$

$\mathrm{h} = X + S$ splits by time-reversal parity (next slide)

Exchange vs. spin–orbit: time-reversal split

Claim

In a time-reversal-symmetric (real) orbital gauge, $X = \mathrm{Re}\,\mathrm{h}$ is *time-odd* (exchange) and $S = i\,\mathrm{Im}\,\mathrm{h}$ is *time-even* (SOC).

Proof

$\Theta = i\sigma_y\mathcal{K}$, $\Theta\sigma_c\Theta^{-1} = -\sigma_c$, and $\mathcal{K}$ conjugates the orbital matrices (real basis functions):

$$\Theta(M \otimes \sigma_c)\Theta^{-1} = -M^* \otimes \sigma_c.$$

Hence the time-even part of h satisfies $M^* = -M$ (purely imaginary) and the time-odd part $M^* = M$ (real). Hermiticity of H gives $\mathrm{h}(\mathrm{R})^\dagger = \mathrm{h}(-\mathrm{R})$, so $\mathrm{Re}\,\mathrm{h}$ is symmetric and $i\,\mathrm{Im}\,\mathrm{h}$ antisymmetric: atomic $\xi\mathrm{L}\cdot\mathrm{S}$ has L imaginary-antisymmetric in a real basis. ■

Diagnostic: $\max|\mathrm{Im}\,H_0|$ is stored; spinor MLWFs are only approximately real, which is why the code also checks $E(\mathrm{n}) = E(-\mathrm{n})$ for the MAE.

Assigning Wannier functions to atoms and to the home cell

- Atom of each WF from the `projections` block (robust), checked against `_centres.xyz`.
- A centre close to an *image* $\boldsymbol{\tau}_a + \mathrm{T}_m$ is moved home: $w'_m(\mathrm{r}) = w_m(\mathrm{r} + \mathrm{T}_m)$.

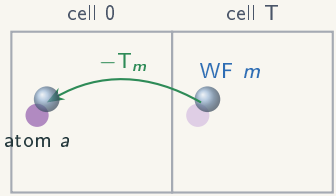
Home-cell shift `_home_shift`

$$H'_{mn}(\mathrm{R}) = H_{mn}(\mathrm{R} + \mathrm{T}_n - \mathrm{T}_m)$$

Proof

$\langle w'_{m\mathbf{0}} | \hat{H} | w'_{n\mathrm{R}} \rangle = \langle w_{m,-\mathrm{T}_m} | \hat{H} | w_{n,\mathrm{R}-\mathrm{T}_n} \rangle$ and translation invariance moves $-\mathrm{T}_m$ to the origin. ■

Residual distance $>$ `wf_centre_tol` (0.6 Å) stops the run: bond-centred MLWFs cannot be attributed to an atom.



Atomic spin–orbit coupling on the Wannier orbitals

Wannier90 real harmonics: $p : (p_z, p_x, p_y)$, $d : (d_{z^2}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{xy})$, f: 7 functions.

L built from the polynomials themselves

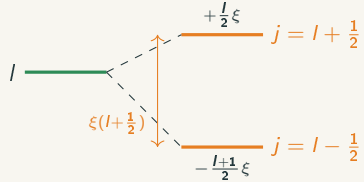
$$L_\alpha\, Y_m = -i(\mathrm{r}\times\nabla)_\alpha\, Y_m = \sum_{m'} Y_{m'}\, C^{\alpha}_{m'm}$$

evaluated on a Gauss–Legendre × uniform-ϕ grid (exact for these polynomials), normalised with exact norms. No tabulated phases ⇒ no sign errors.

Added to the on-site block

$$\xi_l\, \mathrm{L}\cdot\mathrm{S} = \sum_c \underbrace{\frac{\xi_l}{2} L_c}_{S_c} \otimes \sigma_c$$

Checks: $[L_x, L_y] = iL_z$, $\mathrm{L}^2 = l(l{+}1)$, eigenvalues of L·S: $\frac{l}{2}$ and $-\frac{l+1}{2}$.



default ξ table (Cr 35, Fe 55, Pt 600, I 600, Bi 1250 meV); override with `[soc_xi]`.

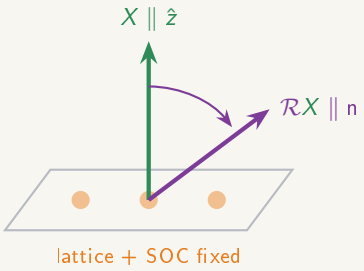
Rotating the magnetisation relative to the lattice

`TBModel.rotated(R)`

$$X_c(\mathrm{R}) \longrightarrow \sum_d \mathcal{R}_{cd}\, X_d(\mathrm{R}), \qquad H_0,\ S \text{ fixed}$$

- rotates the spin direction of the *exchange* field only: the physical rotation of the magnetisation with the crystal and SOC fixed.
- one collinear DFT run + $\xi \mathrm{L}\cdot\mathrm{S}$ gives exact model-level references along x, y, z (or any n).
- a rotation of the *full* spin space would be a gauge change and leave all energies unchanged.

`rotation_to(n)`: Rodrigues rotation taking $\hat{z} \to \mathrm{n}$.



Magnetic force theorem

Counting states: the Lloyd formula

Integrated density of states

$$N(\varepsilon) = N_{\mathrm{tot}} - \frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,\ln\!\left(\varepsilon + i0^{+} - H\right)$$

Proof

For one level E: $\ln(\varepsilon + i0 - E) = \ln|\varepsilon - E| + i\pi\,\theta(E-\varepsilon)$. Summing over eigenvalues, $\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,\ln$ counts the levels *above* ε. ■

Variation under $H \to H + V$

$$\delta N(\varepsilon) = -\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\ln(1 - GV) = \frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\left[GV + \tfrac{1}{2}GVGV + \dots\right], \qquad G = (\varepsilon + i0 - H)^{-1}$$

using $\ln(\varepsilon - H - V) - \ln(\varepsilon - H) = \ln(1 - GV)$ inside the trace.

Grand potential and the force theorem

Frozen-potential grand potential at fixed

$$\Omega = -k_B T \sum_{n} \ln\!\left(1 + e^{-(\varepsilon_n - \mu)/k_B T}\right) = -\int d\varepsilon\, f(\varepsilon)\, N(\varepsilon)$$

Proof

$\frac{d}{d\varepsilon}\left[- k_B T \ln(1 + e^{-(\varepsilon-\mu)/k_B T})\right] = f(\varepsilon)$; integrate $\Omega = \int n(\varepsilon)\,(\cdots)\,d\varepsilon$ by parts with $n = dN/d\varepsilon$. ■

Second-order energy change

$$\delta\Omega = -\frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f(\varepsilon)\,\mathrm{Tr}\left[G V + \tfrac{1}{2} G V G V\right]$$

Force theorem: to second order the self-consistent energy change equals this band (grand-potential) change at frozen potential, because the first-order density response does not change the functional's stationary value. The code computes Ω-curvatures *exactly*; the only approximation is the frozen potential.

The perturbation: a rigid rotation of one site's exchange field

Rotation by ϕ about axis e_a in spin space: $U = e^{-i\phi\, \mathrm{e}_a \cdot \boldsymbol{\sigma}/2}$.

First derivative

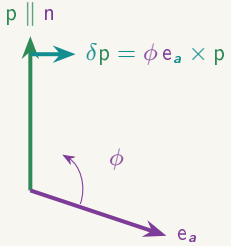
$$\left.\frac{\partial H}{\partial \phi_a}\right|_0 = -\tfrac{i}{2}[\sigma_a, \mathrm{p}\cdot\boldsymbol{\sigma}] = (\mathrm{e}_a \times \mathrm{p})\cdot\boldsymbol{\sigma} \equiv W_a$$

Proof

$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c \Rightarrow -\tfrac{i}{2}[\sigma_a, p_b\sigma_b] = \epsilon_{abc}\, p_b \sigma_c = (\mathrm{e}_a \times \mathrm{p})_c \sigma_c$. ■

Second derivative (symmetrised)

$$\frac{\partial^2 H}{\partial\phi_a \partial\phi_b} = \tfrac{1}{2}\left[\mathrm{e}_a \times (\mathrm{e}_b \times \mathrm{p}) + \mathrm{e}_b \times (\mathrm{e}_a \times \mathrm{p})\right]\cdot\boldsymbol{\sigma}$$



p_i: time-odd on-site field of site i (`onsite_exchange`). Only $\mathrm{e}_a \perp \mathrm{n}$ gives a non-zero W_a.

Two-site and on-site Hessians `magex/engine.py`

$$K_{ab}(i,j,\mathrm{R}) \equiv \frac{\partial^2\Omega}{\partial\phi_{ia}\partial\phi_{jb}} = -\frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f(\varepsilon)\,\mathrm{Tr}\left[W_{ia}\,G_{ij}(\mathrm{R})\,W_{jb}\,G_{ji}(-\mathrm{R})\right]$$

$$\frac{\partial^2\Omega}{\partial\phi_{ia}\partial\phi_{ib}} = \tfrac{1}{2}\left[K_{ab}(i,i,0)+K_{ba}(i,i,0)\right] + T_{ab}(i), \qquad T_{ab}(i) = \mathrm{Tr}\Big[\rho_{ii}\,\frac{\partial^2 H_i}{\partial\phi_a\partial\phi_b}\Big]$$

Why the cross term has no factor $\frac{1}{2}$

With $V = \phi_{ia}W_{ia} + \phi_{jb}W_{jb}$, the term $\frac{1}{2}\,\mathrm{Tr}[G\,V\,G\,V]$ contains $\phi_{ia}\phi_{jb}\,\mathrm{Tr}[G\,W_{ia}\,G\,W_{jb}]$ twice $\times\frac{1}{2}$ (cyclicity). The first-order term with $\partial^2 H$ gives T via $\rho = -\frac{1}{\pi}\,\mathrm{Im}\int f\,G$. ■

$G_{ij}(\mathrm{R})$: block between site i in cell 0 and site j in cell R. All 9 axis pairs (a,b) are computed; those with e ∥ n vanish.

Spectral form (no matrix inversion per z)

$$G(\mathrm{k}, z) = \Psi_{\mathrm{k}}\, \mathrm{diag}\Big(\frac{1}{z - \varepsilon_{nk}}\Big) \Psi_{\mathrm{k}}^{\dagger}$$

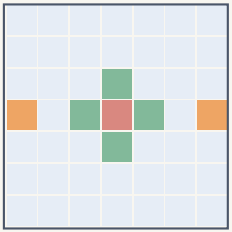
To real space by FFT

$$G(\mathrm{R}, z) = \frac{1}{N_k} \sum_{\mathrm{k}} G(\mathrm{k}, z)\, e^{-2\pi i \mathrm{k}\cdot\mathrm{R}} = \texttt{fftn}/N_k$$

$$G(-\mathrm{R}) = \texttt{roll(flip}(G), 1)$$

- only rows/columns of the spin sites are kept: Ψ_{sel} of shape $(N_k, N_{\mathrm{sel}}, N_b)$
- Γ-centred mesh $\Rightarrow$ R lives on the $N_1 \times N_2 \times N_3$ supercell
- test: K from the k-mesh $=$ Γ-only explicit supercell to 9×10^{-16}

orange: R and R $-$ *N*a tie at the boundary $\Rightarrow$ shared



7×7 supercell of R

Energy integration

Continued-fraction expansion

$$f(x) = \tfrac{1}{2} - \sum_{p=1}^{N_P} R_p \left[\tfrac{1}{x - i z_p} + \tfrac{1}{x + i z_p} \right]$$

$z_p = 1/\lambda_p$ from the eigenvalues $\lambda_p > 0$ of the tridiagonal B, $b_n = \frac{1}{2\sqrt{(2n-1)(2n+1)}}$; $R_p = |v_{1p}|^2 / 4\lambda_p^2$.

Contour result

$$-\frac{1}{\pi} \operatorname{Im} \int f \, F(\varepsilon^+) \, d\varepsilon = 2 k_B T \sum_p R_p \operatorname{Re} F(\mu + i z_p k_B T)$$

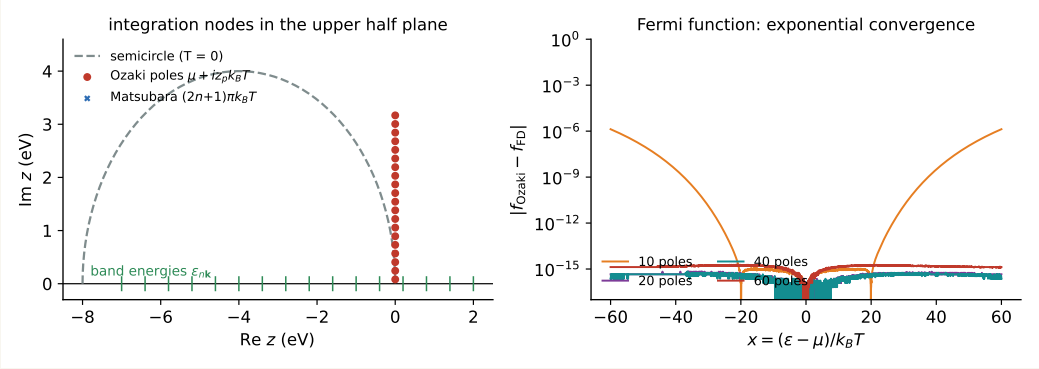
Proof

F analytic in the UHP and $F \sim z^{-2}$: close the real line in the UHP. $\int F/2 = 0$; only the poles of f at $\mu + i z_p k_B T$ (residue $-R_p k_B T$) contribute: $\int fF = -2\pi i \, k_B T \sum_p R_p F(z_p)$. Take $-\frac{1}{\pi} \operatorname{Im}$. ■

Why $F \sim z^{-2}$: every perturbation is a Pauli matrix, $\operatorname{Tr}_\sigma \sigma_c = 0$, so the $1/z$ term $\operatorname{Tr} V/z$ vanishes.

Gain: no $E_{\min}$, 60 poles for 30 eV at 300 K, $\mu + i\pi k_B T$ smooths k-sums in metals.

Integration nodes and convergence



Left: Ozaki poles sit on the line $\mathrm{Re}\, z = \mu$, sparser than Matsubara frequencies and far from all bands. Right: error of the Fermi function vs. number of poles; `ozaki_npoles` picks the smallest N_p reaching 10^{-10} over the band width. The $T = 0$ semicircle (Gauss–Legendre in angle) is kept as an option.

Rotational invariance

Exact statement (no SOC)

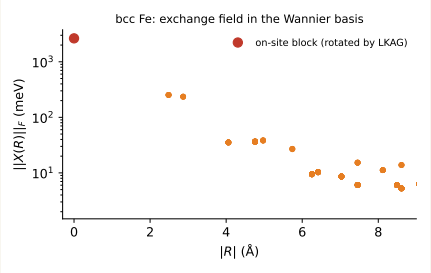
A uniform rotation of *all* spins is a symmetry $\Rightarrow \Omega$ is constant along it:

$$\sum_{j,\mathrm{R}} K_{aa}(i,j,\mathrm{R}) + T_{aa}(i) = 0 \qquad \forall\, i$$

Proof

Rotate every site by the same ϕ: $\Omega(\phi) - \Omega(0) = \frac{1}{2}\phi^2 \sum_{ij\mathrm{R}} \partial^2\Omega = 0$ only if the rotated operators add up to the generator of a *global* rotation, $\sum_i \delta H_i = (\mathrm{e}_a \times X) \cdot \boldsymbol{\sigma}$ for the full $X(\mathrm{R})$. ■

LKAG/TB2J rotate only $\mathrm{p}_i = X_{ii}(0)$. In Wannier bases the exchange field also lives on inter-atomic elements $X_{ij}(\mathrm{R})$, which then never rotate.



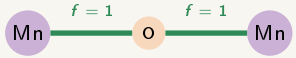
Partitioning the exchange field among atoms

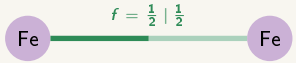
Weighted partition

$$X = \sum_i X^{(i)}, \qquad X^{(i)} = \Pi_i X_L + X_L^\dagger \Pi_i$$

$$(X_L)_{mn}(\mathrm{R}) = f_{a(m)a(n)}\, X_{mn}(\mathrm{R}), \quad f_{ab} = \frac{w_a}{w_a + w_b}$$

$w_a = \|\mathrm{p}_a\|$; $f_{aa} = \frac{1}{2}$. Π_i projects on the orbitals of atom i in cell 0.





magnetic atoms carry the spin-dependent hopping to unpolarised neighbours

$\sum_i X^{(i)} = X$

$\sum_i \Pi_i = 1$ over all atoms and cells, so $\sum_i X^{(i)} = X_L + X_L^\dagger$. Element-wise $(X_L + X_L^\dagger)_{mn} = (f_{ab} + f_{ba}) X_{mn} = X_{mn}$ using $X^\dagger = X$. ■

Take-away: rotating every $X^{(i)}$ by the same angle is an exact global spin rotation, so the sum rule holds identically.

Perturbation of site i

$$W_{ia} = \Pi_i Y_a + Y_a{}^\dagger \Pi_i, \qquad Y_a = (\mathrm{e}_a \times X_L)\cdot\boldsymbol{\sigma}$$

Trace (`engine_partition`)

$$\begin{aligned}\mathrm{Tr}[W_{ia} G W_{jb} G] = {} & \mathrm{Tr}\left[(Y_a G)_{ij}(Y_b G)_{ji}\right] + \mathrm{Tr}\left[(Y_a G Y_b{}^\dagger)_{ij} G_{ji}\right] \\ & + \mathrm{Tr}\left[G_{ij}(Y_b G Y_a{}^\dagger)_{ji}\right] + \mathrm{Tr}\left[(G Y_b{}^\dagger)_{ij}(G Y_a{}^\dagger)_{ji}\right]\end{aligned}$$

Proof

Expand the product of two sums; move each Π by cyclicity, e.g. $\mathrm{Tr}[Y_a{}^\dagger \Pi_i G Y_b{}^\dagger \Pi_j G] = \mathrm{Tr}[\Pi_i (G Y_b{}^\dagger)\Pi_j (G Y_a{}^\dagger)]$; $\Pi_i \cdots \Pi_j$ selects the (i,j) block. ■

The on-site first-order term becomes $T_{ab}(i) = \mathrm{Tr}_i[(Z_{ab}\rho)_{ii} + (\rho Z_{ab}{}^\dagger)_{ii}]$, $Z_{ab} = \frac{1}{2}[\mathrm{e}_a \times (\mathrm{e}_b \times X_L) + (a \leftrightarrow b)]\cdot\boldsymbol{\sigma}$.

Evaluating the four terms efficiently

Precompute once per k (independent of z):

$$U_a(\mathrm{k}) = Y_a(\mathrm{k})[\mathrm{sel},:]\,\Psi_{\mathrm{k}}, \qquad Q_{ab}(\mathrm{k}) = Z_{ab}(\mathrm{k})[\mathrm{sel},:]\,\Psi_{\mathrm{k}}$$

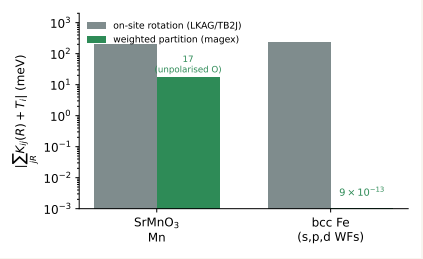
Then every block has the form $A\,\mathrm{diag}(g)\,B^\dagger$ with $g_n = 1/(z-\varepsilon_{n\mathrm{k}})$:

$$G = \Psi_{\mathrm{s}} g \Psi_{\mathrm{s}}^\dagger, \qquad Y_a G = U_a g \Psi_{\mathrm{s}}^\dagger,$$
$$G Y_b^\dagger = \Psi_{\mathrm{s}} g U_b{}^\dagger, \qquad Y_a G Y_b^\dagger = U_a g U_b{}^\dagger$$

16 batched `matmuls` + 16 FFTs per energy node.

Exact checks

- `partition="onsite"` reproduces the fast engine to 4×10^{-16}
- weighted Hessian vs. finite differences of the partitioned rotation: 1.3×10^{-7}
- sum rule: 4×10^{-15}



From Hessians to spin-model parameters

Mapping used for every pair `magex/heisenberg.py`

$$K_{ab}(i,j,\mathrm{R}) = -2\,(\mathrm{e}_a \times \mathrm{n}_i)\cdot J_{ij}(\mathrm{R})\cdot(\mathrm{e}_b \times \mathrm{n}_j)$$

Proof

Rotating $\mathrm{e}_i = \mathcal{R}(\phi_i)\mathrm{n}_i$ gives $\delta\mathrm{e}_i = \phi_{ia}\,\mathrm{e}_a \times \mathrm{n}_i + O(\phi^2)$. The pair (i,j) enters E as $-\mathrm{e}_i\cdot J_{ij}\mathrm{e}_j - \mathrm{e}_j\cdot J_{ji}\mathrm{e}_i = -2\,\mathrm{e}_i\cdot J_{ij}\cdot\mathrm{e}_j$, whose mixed derivative is $-2(\mathrm{e}_a\times\mathrm{n}_i)\cdot J_{ij}\cdot(\mathrm{e}_b\times\mathrm{n}_j)$. Second-order terms in $\delta\mathrm{e}_i$ multiply n_j and do not contribute to $\partial_{\phi_i}\partial_{\phi_j}$. ■

- collinear reference: $\mathrm{n}_i = s_i\hat{\mathrm{d}}$ with $s_i = \mathrm{sign}(\mathrm{M}_i\cdot\hat{\mathrm{d}})$ (AFM handled automatically)
- isotropic, no SOC: $K_{aa} = -2s_is_j\,J_{ij}$ for $a \perp \hat{\mathrm{d}}$; equivalent to LKAG $J = \frac{1}{4\pi}\,\mathrm{Im}\int \mathrm{Tr}[\Delta_i G_{ij}^{\uparrow}\Delta_j G_{ji}^{\downarrow}]$ averaged over $i \leftrightarrow j$

Three references from one Hamiltonian

Each reference $\hat{\mathrm{d}}$ gives 4 equations (axes $\perp \hat{\mathrm{d}}$):

- $\hat{\mathrm{d}} = \hat{z}$: $J_{xx}, J_{xy}, J_{yx}, J_{yy}$
- $\hat{\mathrm{d}} = \hat{x}$: $J_{yy}, J_{yz}, J_{zy}, J_{zz}$
- $\hat{\mathrm{d}} = \hat{y}$: $J_{zz}, J_{zx}, J_{xz}, J_{xx}$

12 equations, 9 unknowns $\Rightarrow$ least squares:

Joint fit

$$\min_{J_{ij}} \sum_{\hat{\mathrm{d}}} \sum_{a,b} \left| K^{(\hat{\mathrm{d}})}_{ab} + 2(\mathrm{e}_a \times \mathrm{n}_i) \cdot J_{ij} \cdot (\mathrm{e}_b \times \mathrm{n}_j) \right|^2$$

residual = **reference dependence** (printed per pair).

	x	y	z
x	z, y	z	y
y	z	z, x	x
z	y	x	x, y

references that see $J_{\alpha\beta}$
(diagonal: measured twice)

Take-away: one collinear DFT run + $\xi \mathrm{L} \cdot \mathrm{S}$ $\rightarrow$ J^{iso}, J^{ani} and D for all pairs; no three-run merge, no inter-run Wannier noise.

Single-ion anisotropy from the on-site Hessian

Energy of site i alone rotated (all other spins frozen)

$$E_i(\mathrm{e}) = \mathrm{e}\cdot A_i\cdot\mathrm{e} + \mathrm{c}\cdot\mathrm{e}, \qquad \mathrm{c} = -2\sum_{(j,\mathrm{R})\neq(i,0)} J_{ij}(\mathrm{R})\,\mathrm{n}_j$$

Its Hessian (`build_results`)

$$\frac{\partial^2 E_i}{\partial\phi_a\partial\phi_b} = 2(\mathrm{e}_a\times\mathrm{n})\cdot A_i\cdot(\mathrm{e}_b\times\mathrm{n}) + 2\,\mathrm{n}\cdot A_i\cdot\mathrm{S}_{ab} + \mathrm{c}\cdot\mathrm{S}_{ab} + \delta_i\,(\mathrm{e}_a\times\mathrm{n})\cdot(\mathrm{e}_b\times\mathrm{n})$$

$$\mathrm{S}_{ab} = \tfrac{1}{2}\big[\mathrm{e}_a\times(\mathrm{e}_b\times\mathrm{n}) + \mathrm{e}_b\times(\mathrm{e}_a\times\mathrm{n})\big]$$

- equated to $\frac{1}{2}(K_{ab}+K_{ba})(i,i,0)+T_{ab}(i)$ for all three references; unknowns: 5 components of traceless A_i and an isotropic offset δ_i
- images of i in other cells are *different* (frozen) sites: they enter linearly through c
- δ_i absorbs any sum-rule violation; without SOC $A_i = 0$ follows automatically

Magnetocrystalline anisotropy and its cross-check

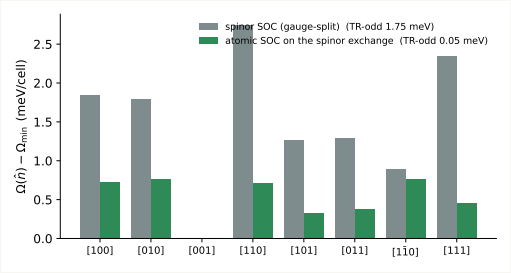
Band MAE (`grand_potential_directions`)

$$\Omega(\mathrm{n}) = -\frac{k_B T}{N_k} \sum_{n\mathbf{k}} \ln\left(1 + e^{-(\varepsilon_{n\mathbf{k}}(\mathrm{n}) - \mu)/k_B T}\right)$$

8 directions and their negatives, dense `mae_kmesh`.

- **time-reversal check**: $\max|\Omega(\mathrm{n}) - \Omega(-\mathrm{n})|/2$ must vanish
- even part fitted to $\mathrm{n} \cdot A_{\mathrm{band}} \cdot \mathrm{n}$
- compared with the Heisenberg prediction

$$E_H(\mathrm{n}) = \sum_i \mathrm{n} \cdot A_i \cdot \mathrm{n} - \sum_{ij\mathrm{R}} s_i s_j \, \mathrm{n} \cdot J_{ij} \cdot \mathrm{n}$$



CrI_3: with gauge-split spinor SOC the TR-odd error is 1.75 meV; with atomic SOC 0.05 meV, easy axis z, $K_{\mathrm{eff}} = 0.71$ (band) vs. 0.78 meV/cell (Heisenberg).

Symmetrisation and supercell aliasing

Group average over the space group (spglib)

$$J(p) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathcal{R}_g{}^{T}\, J(g\,p)\, \mathcal{R}_g, \qquad A_i \text{ likewise}$$

Proof

J couples two axial vectors: $\mathrm{e} \to \det(\mathcal{R})\mathcal{R}\mathrm{e}$ for both, $\det^2 = 1$, so $J(gp) = \mathcal{R} J(p) \mathcal{R}^T$; time reversal flips both spins and leaves J unchanged. ■

Atoms are typed by species and |M|; pairs are looked up by their FFT index $\nu = \mathrm{R} \bmod N$; the size of the change is printed as the Wannier-noise estimate.

Minimal images. $\mathrm{R} \equiv \nu \pmod N$ chosen to minimise $|\boldsymbol{\tau}_j + \mathrm{R} - \boldsymbol{\tau}_i|$ (`min_image`). At the supercell boundary several images tie: the coupling is *shared*, J/n_{img} on each (`all_images`), keeping $h(\mathrm{k})$ Hermitian.

$SrMnO_3$: before sharing, $h(\mathrm{k})$ was non-Hermitian and LSWT reported a spurious instability.

Beyond the plain force theorem

Bruno renormalisation: rotating moments, not fields

Let θ_i be moment angles and ϕ_i field angles; linear response $\theta = X\phi$.

$$K = K_b + D, \qquad D_i = T_i, \qquad K_b : \text{ bubble incl. on-site}$$

Derivation of $\tilde{K} = -DK_b{}^{-1}D - D$

Hellmann–Feynman: $\partial\Omega/\partial\phi = D(\phi - \theta)$, hence $K = D(1 - X)$, $X = -D^{-1}K_b$. A constraining field fixes the moments at θ while the exchange field follows them; total field $\phi_{\mathrm{tot}} = X^{-1}\theta$. The constraining torque is $\partial E/\partial\theta = D(\phi_{\mathrm{tot}} - \theta) = D(X^{-1} - 1)\theta$, so $\tilde{K} = D(X^{-1} - 1) = -DK_b{}^{-1}D - D$. ∎

One FM sublattice

$$\tilde{\omega}(\mathrm{q}) = \frac{\omega(\mathrm{q})}{1 - \omega(\mathrm{q})/\Delta}$$

$\tilde{K} = K + KD^{-1}K + \ldots$ raises spin-wave energies comparable to the exchange splitting Δ.

Implemented per q in the magnetic subspace (`renorm.py`); D from the sum rule so that $\tilde{K}(0)$ keeps the Goldstone zero mode.

Ligand downfolding

Include spin-polarised ligands (O, I, Se . . .) as rotatable sites, then let them relax:

Schur complement per q

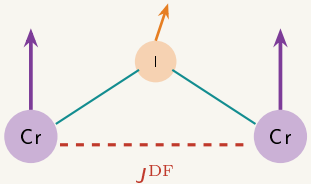
$$K_{\mathrm{eff}}(\mathbf{q}) = K_{MM} - K_{ML} K_{LL}^{-1} K_{LM}$$

Proof

$E = \frac{1}{2}\phi_M^\dagger K_{MM}\phi_M + \mathrm{Re}\,\phi_M^\dagger K_{ML}\phi_L + \frac{1}{2}\phi_L^\dagger K_{LL}\phi_L$;
$\partial E/\partial\phi_L^* = 0$ gives $\phi_L = -K_{LL}^{-1}K_{LM}\phi_M$; substitute. ■

$$J_{ij}^{\mathrm{DF}}(\mathbf{R}) = -\frac{1}{2 s_i s_j}\,\mathrm{FFT}^{-1}\big[K_{\mathrm{eff}}(\mathbf{q})\big]_{ij}$$

Ligands with $\|\mathbf{p}\| < 5$ meV are dropped automatically.



induced ligand moment follows the metals adiabatically

Finite angles: exact Lloyd energies

Local perturbation of a site set S

$$\Delta\Omega_S = \frac{1}{\pi}\,\mathrm{Im}\int d\varepsilon\, f\ \ln\det\left(1 - G_{SS}V_{SS}\right) \;=\; -2k_B T \sum_p R_p\ \ln\left|\det(1 - GV)\right|_{z_p}$$

$V = (\mathcal{R}(\theta)X^{(i)} - X^{(i)})\cdot\boldsymbol{\sigma}$, finite. Only $\ln|\det|$ is needed (Ozaki weights are real): no branch cuts.

Non-local V (weighted partition): Sylvester

$V = \Pi_i A + A^\dagger \Pi_i = LM$, $L = [\Pi_i\ A^\dagger]$, $M = [A; \Pi_i]$. Then $\det(1 - GLM) = \det(1 - MGL)$, a $2n_i \times 2n_i$ matrix with blocks $(AG)_{ii}$, $(AGA^\dagger)_{ii}$, G_{ii}, $(GA^\dagger)_{ii}$; for a pair, the 4×4 block version with phases $e^{\mp 2\pi i \mathbf{k}\cdot\mathbf{R}}$. ■

Pair interaction energy: $E_{\mathrm{int}} = \Delta\Omega_{ij} - \Delta\Omega_i - \Delta\Omega_j$. Exact vs. brute-force Ω: 2.5×10^{-11} eV.

Bilinear and biquadratic parts of the pair energy

Rotate i by $-\theta/2$ and j by $+\theta/2$; $c = \cos(\theta/2)$, $s = s_i s_j$.

Model $E = -\sum_{i \neq j} \left[\; \mathrm{e}_i \cdot \mathrm{e}_j + \; (\mathrm{e}_i \cdot \mathrm{e}_j)^2 \right]$

$$E_{\mathrm{int}}(\theta) = 4s\, J'\, c(1-c) - 2B \cos\theta(\cos\theta - 1)$$

$$J_{\mathrm{small}} = J' + 2s\, B$$

Proof

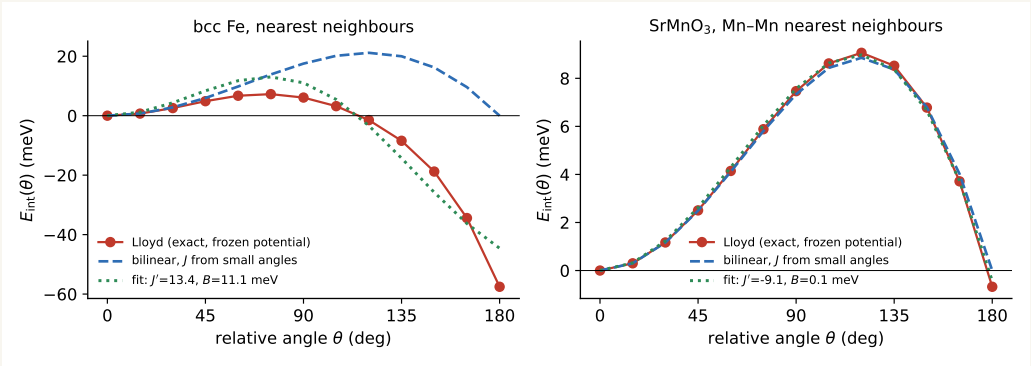
$E_{\mathrm{int}} = -2J'(\mathrm{e}_i - \mathrm{n}_i)\cdot(\mathrm{e}_j - \mathrm{n}_j) + \ldots$; with $\mathrm{e}_i - \mathrm{n}_i = (-\sin\frac{\theta}{2}, 0, c-1)$, $\mathrm{e}_j - \mathrm{n}_j = s(\sin\frac{\theta}{2}, 0, c-1)$ the dot product is $-2sc(1-c)$. The B term: $(\cos^2\theta - 2c^2 + 1) = \cos\theta(\cos\theta - 1)$. Small θ: $E \simeq \frac{\theta^2}{2}(sJ' + 2B)$. ■



reference $\mathrm{n}_i, \mathrm{n}_j$ (faint)

The code also draws the *tensor* bilinear curve $-2(\mathrm{e}_i - \mathrm{n}_i)\cdot J_{ij} \cdot (\mathrm{e}_j - \mathrm{n}_j)$ and reports $\max|E_{\mathrm{Lloyd}} - E_{\mathrm{bil}}|$ as the Heisenberg error.

Lloyd results: how Heisenberg-like are the bonds?



$SrMnO_3$ (localised Mn^{4+}): the bilinear curve from small angles reproduces the exact pair energy over the full range. bcc Fe: strongly angle-dependent nearest-neighbour coupling, a large effective biquadratic term, in line with Szilva *et al.*, PRL 111, 127204 (2013). Small-angle J remain the right input for low-temperature magnons.

Spin waves

Local frames: n_i (local z), $\mathsf{u}_i = \mathsf{x}_i + i\mathsf{y}_i$; spins $S_i = |\mathsf{M}_i|/2$; $\tilde{J}_{ij} = -J_{ij}/S_iS_j$, $\tilde{J}_{ii}(0) = A_i/S_i^2$.

Holstein–Primakoff

$$\mathsf{S}_i \simeq \sqrt{S_i/2}\,\left(\bar{\mathsf{u}}_i b_i + \mathsf{u}_i b_i^\dagger\right) + \mathsf{n}_i(S_i - b_i^\dagger b_i)$$

Bosonic Bogoliubov–de Gennes matrix

$$A_{ij}(\mathsf{k}) = \sqrt{S_iS_j}\;\mathsf{u}_i^T \tilde{J}_{ij}(\mathsf{k})\bar{\mathsf{u}}_j - 2\delta_{ij}\sum_l S_l\;\mathsf{n}_i\cdot\tilde{J}_{il}(0)\cdot\mathsf{n}_l$$

$$B_{ij}(\mathsf{k}) = \sqrt{S_iS_j}\;\mathsf{u}_i^T \tilde{J}_{ij}(\mathsf{k})\mathsf{u}_j, \qquad h(\mathsf{k}) = \begin{pmatrix} A(\mathsf{k}) & B(\mathsf{k}) \\ B(\mathsf{k})^\dagger & A(-\mathsf{k})^T \end{pmatrix}$$

Energies: positive eigenvalues of $\mathrm{diag}(1,-1)\,h(\mathsf{k})$ (paraunitary). Complex eigenvalues $\Rightarrow$ reference not a minimum (warning).

LSWT: checks against closed forms

FM: $\omega(\mathrm{q}) = \frac{4}{M}\big[J(0) - J(\mathrm{q})\big]$

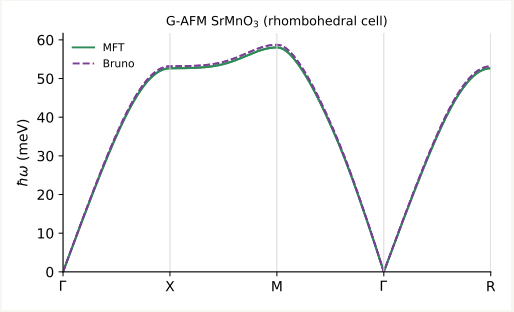
$\mathrm{u}\cdot\bar{\mathrm{u}} = 2$, $\mathrm{u}\cdot\mathrm{u} = 0 \Rightarrow B = 0$ and $A = 2S\big[\tilde{J}(\mathrm{q}) - \tilde{J}(0)\big] = \frac{2}{S}\big[J(0) - J(\mathrm{q})\big]$ with $S = M/2$; the same as the classical precession $\hbar S\dot{\mathrm{e}} = -\mathrm{e} \times \partial E/\partial \mathrm{e}$. ■

AFM chain

$\omega = -\frac{4J}{S}|\sin \pi q|$ reproduced to 10^{-8}.

Easy-axis gap

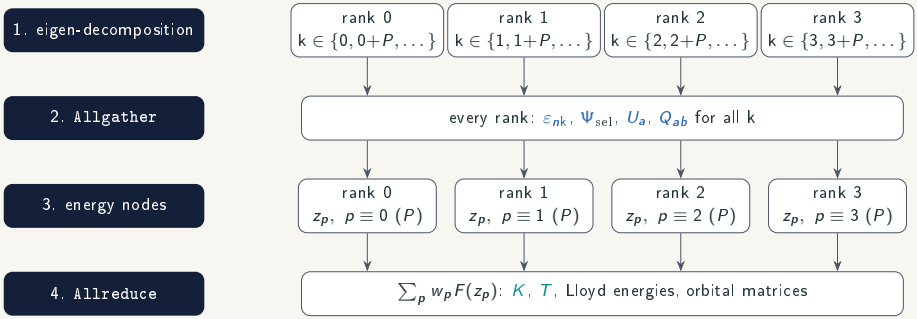
$A = -K\hat{z}\hat{z}$: $\omega(0) = 2K/S$, exact.



G-type $SrMnO_3$: two degenerate Goldstone branches, linear at Γ; Bruno raises the zone-boundary energies slightly (insulator, $\omega \ll \Delta$).

Parallel implementation

MPI layout



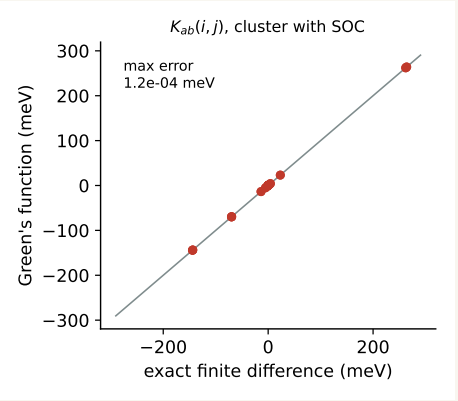
task	cost per energy node	parallel axis
fast engine (on-site)	$N_k N_{\mathrm{sel}}^2 N_b$ + FFT	z
weighted engine	$16 \times N_k N_{\mathrm{sel}}^2 N_b$ + 16 FFTs	z
Lloyd (per pair, angle)	$12 \times N_k n^2 N_b$	z
MAE	N_k diagonalisations per direction	k

$SrMnO_3$, 5^3: 34.6 s $\rightarrow$ 18.8 s (2 ranks)
results identical to 10^{-14}
`OMP_NUM_THREADS=1`, 1 rank/core

Validation and results

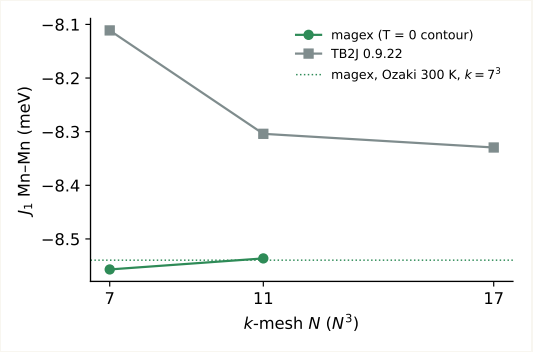
Exactness tests

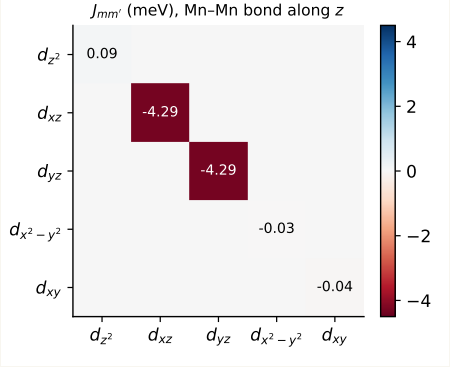
test (`tests/`)	max error
GF Hessian vs. finite differences, cluster with SOC, AFM, spin-dependent hopping	1.5×10^{-7}
weighted partition vs. finite differences	1.3×10^{-7}
k-mesh/FFT vs. explicit supercell	9×10^{-16}
Lloyd vs. exact Ω (both partitions)	2.5×10^{-11}
Goldstone sum rule (weighted)	4×10^{-15}
Heisenberg round trip (J, A)	7×10^{-16}
LSWT: FM, AFM chain, gap	analytic
real $SrMnO_3$: $\sum_R K$ vs. FD	$-66.571 / -66.565$



finite-difference error ($h = 10^{-3}$) dominates the residual.

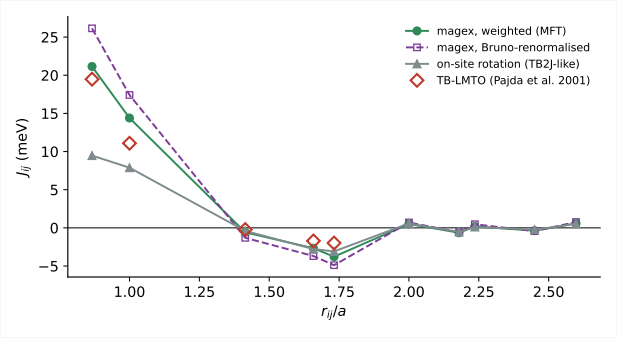
$SrMnO_3$: comparison with TB2J





- same Wannier data, on-site partition: magex $J_1 = -8.54$ meV (converged at 7^3), TB2J 0.9.22 -8.33 meV at 17^3
- t_{2g} channel identical (-4.291 meV for d_{xz}, d_{yz}); the difference sits in the e_g channel
- magex matches brute-force energy differences of the same Hamiltonian to 10^{-4}

bcc Fe: the effect of rotational invariance



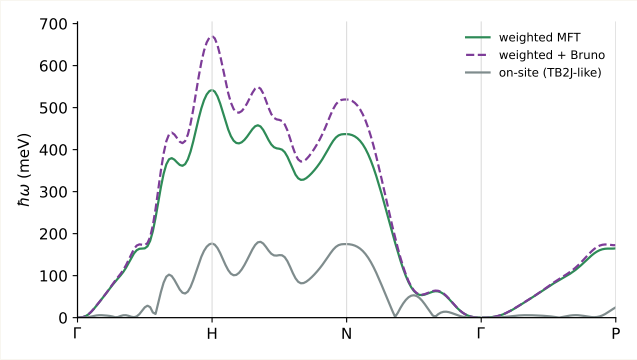
	J_1 (meV)	T_{MF} (K)
on-site (TB2J-like)	9.5	441
weighted, MFT	21.2	1427
weighted, Bruno	26.1	1660
TB-LMTO (Pajda)	19.5	1414
experiment	–	1043

$k_B T_{\mathrm{MF}} = \frac{2}{3} \sum_j J_{0j}$

QE + Wannier90 (s,p,d), 13^3, 2 MPI ranks, 2.7 min.

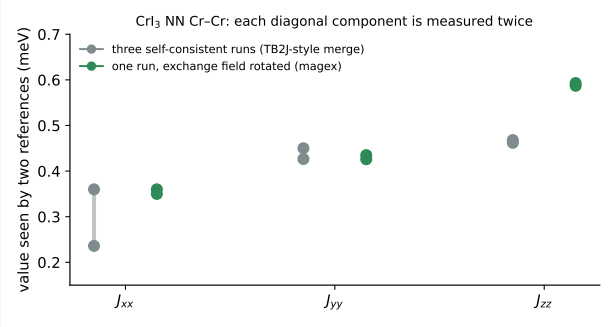
Take-away: rotating the inter-atomic exchange field with the moment turns an unphysical Wannier result into LMTO-quality exchange.

bcc Fe: spin waves



- weighted MFT: maximum 542 meV at H, typical of LKAG Fe results
- Bruno: zone-boundary energies up by $\sim$20% where ω approaches $\Delta \approx 2$ eV
- on-site rotation: soft spectrum, consistent with $T_{\mathrm{MF}} = 441$ K
- the residual wiggles reflect the finite 13^3 real-space range of $J(\mathrm{R})$

CrI_3: one run vs. three runs



- TB2J example data: spinor VASP runs with spins along x, y, z (rotated lattices)
- every diagonal component is measured by two references
- three separately Wannierised runs disagree by up to 35% (J_{xx})
- one run with X rotated agrees with itself to 3%
- TR partner test ($n \to -n$): 1–5%

Take-away: the single-run approach removes inter-run Wannier noise; for spinor input check the TR line, or use collinear + atomic SOC.

Using the code

Input and output

```
[input]
win = "wannier90.up.win"
hr_up = "wannier90.up_hr.dat"
hr_dn = "wannier90.dn_hr.dat"
centres = "wannier90.up_centres.xyz"
efermi = 4.327
[magnetic]
elements = ["Mn"];  ligands = ["O"]
[soc_xi]                    # optional atomic SOC
"Mn:d" = 0.045
[calc]
kmesh = [9, 9, 9]
partition = "weighted"    # or "onsite"
refs = "auto"             # x, y, z with SOC
mae = true
[magnon]
path = "GXMGR"
```

```
mpirun -n 40 magex run in.toml
```

exchange.out: moments and charges, sum rule, J (MFT, Bruno, downfolded), D, J^{ani} with residuals, A_i and easy axes, Lloyd table, MAE with TR and Heisenberg checks, orbital-resolved J, timings.

results.json: all pairs in the supercell, full tensors.

pairs.csv, **magnon_band_*.dat/png**.

Limitations: frozen potential; collinear references; spinor input needs a near-real gauge; Bruno/downfolding act on the isotropic channel; no local-axis projections.

Summary

Theory

- exact force-theorem Hessians of Ω for rotations of the exchange field
- weighted partition $\Rightarrow$ exact rotational invariance
- one Hamiltonian, three references $\Rightarrow$ J, D, J^{ani}, A
- Ozaki poles, Bruno, downfolding, Lloyd, LSWT

Results

- machine-precision tests of every formula
- bcc Fe: $J_1 = 21.2$ meV, $T_{\mathrm{MF}} = 1427$ K (LMTO 19.5, 1414 K)
- $SrMnO_3$ Heisenberg-like, Fe strongly non-Heisenberg at large angles
- CrI_3: single-run tensors self-consistent to 3%

Next: VI_3 and Pt/VI_3 magnons with SIA; CrSb (altermagnet) and interface exchange; self-consistent constrained references; GPU port of the weighted engine.